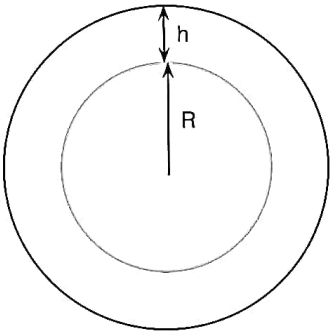
QPT-3 Solution

1. **Ans: (c)** $h = 7R$

Solution:

The equation for escape velocity is $v_e = \sqrt{\frac{2GM}{R}} \quad \ldots(1)$



If satellite is moving in a circular orbit then, Gravitational force = Centripetal force

$\frac{GMm}{(R+h)^2} = \frac{mv^2}{R+h}$

$v = \sqrt{\frac{GM}{(R+h)}}$

It is given that,

$v_c = \frac{1}{4} v_e$

$\sqrt{\frac{GM}{(R+h)}} = \frac{1}{4}\sqrt{\frac{2GM}{R}}$

On taking square on both sides, we get,

$\frac{GM}{R+h} = \frac{1}{16}\frac{2GM}{R}$

$\frac{GM}{R+h} = \frac{GM}{8R}$

$R + h = 8R$

$h = 8R - R$

$h = 7R$

2. **Ans: (b)** $\frac{K}{2}$

Solution: According to the principle of conservation of angular momentum, $I\omega$ = constant

As I is doubled, ω becomes half

Now K.E. of rotation, $K = \frac{1}{2} I\omega^2$

Since I is doubled and ω is halved, K.E. will become half i.e. $\frac{K}{2}$.

3. **Ans: (a)** $\left(-\frac{R}{50}, \frac{21R}{80}\right)$

Solution: Let, $M_2 = M, M_1 = 4M$

$$x_{cm} = \frac{4M(O) - M\left(\frac{R}{10}\right)}{4M + M}$$

$$\Rightarrow x_{cm} = -\frac{R}{50}$$

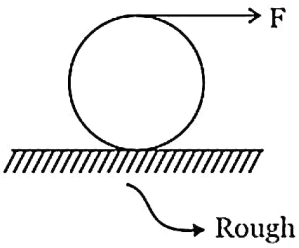
$$y_{cm} = \frac{4M\left(\frac{3R}{8}\right) - M\left(\frac{3R}{16}\right)}{4M + M}$$

$$\Rightarrow y_{cm} = \frac{21R}{80}$$

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4. **Ans: (a)** work done by force F when the center moves a distance S is $2FS$

Solution:



When the centre moves a distance S, the topmost point displaces by $2S$. The work done is by the force is

$W = 2FS$

5. **Ans: (c)** 3 km s^{-1}

Solution:

$$g = \frac{GM}{R^2} = \frac{G}{R^2}\frac{4}{3}\pi R^3\rho = \frac{4}{3}\pi GR\rho \text{ i.e. } g \propto R\rho$$

$$\frac{g_P}{g_E} = \frac{\sqrt{6}g_E}{11g_E} = \frac{\sqrt{6}}{11} \text{ and } \frac{\rho_E}{\rho_P} = \frac{3}{2} \text{ and } \frac{g_P}{g_E} = \frac{R_P\rho_P}{R_E\rho_E}$$

$$\therefore \frac{R_P}{R_E} = \frac{g_P\rho_E}{g_E\rho_P} = \frac{\sqrt{6}}{11} \times \frac{3}{2}$$

$\therefore$ Escape velocity $v = \sqrt{2gR}$

$$\therefore \frac{v_P}{v_E} = \sqrt{\frac{g_P}{g_E} \times \frac{R_P}{R_E}} = \sqrt{\frac{\sqrt{6}}{11} \times \frac{\sqrt{6}}{11} \times \frac{3}{2}} = \frac{3}{11}$$

$$\therefore v_P = \frac{3}{11} \times v_E = \frac{3}{11} \times 11 = 3 \text{ km s}^{-1}$$

6. **Ans: (c)** Speed of all the particles lying on the curved surface of a cylinder whose axis coincides with the axis of rotation is same

Solution: All points in the body, in plane perpendicular to the axis of rotation revolve in connection circles. All points lying on circle of same radius have same speed (and also same magnitude of acceleration) but different directions of velocity (also different directions of acceleration). Hence there cannot be two points in the given plane with same velocity or with same acceleration. As mentioned above, points lying on circle of same radius have same speed.
Angular speed of body at any instant w.r.t. any point on body is same by definition.

7. **Ans: (a)** If both assertion and reason are true and reason is the correct explanation of assertion.

Solution: According to Kepler's law, $\frac{dA}{dt} = \text{constant}$ As, $\frac{dA}{dt} = \frac{1}{2}r^2\frac{d\theta}{dt} = \frac{1}{2}r^2\omega$ $\therefore \frac{mr^2\omega}{2m} = \frac{L}{2m} = \text{constant}$

$\therefore L = \text{constant}$

8. **Ans: (c)** $\frac{mg\sin\theta}{2}$

Solution: Minimum force is required when this force is applied perpendicular to the rod balancing torque about the hinge

$mg\frac{l}{2}\sin\theta = Fl$

$\Rightarrow \quad F = \frac{mg\sin\theta}{2}$

9. **Ans: (c)** $\frac{\omega^2R^2}{2g}$

Solution: The value of acceleration due to gravity at the equator is $g_E = g - R\omega^2$ and the acceleration due to gravity at a height h above the pole is $g_h = \left(1 - \frac{2h}{R}\right)g$. It is given that

$\because mg_E = mg_h$

$\therefore g_E = g_h$

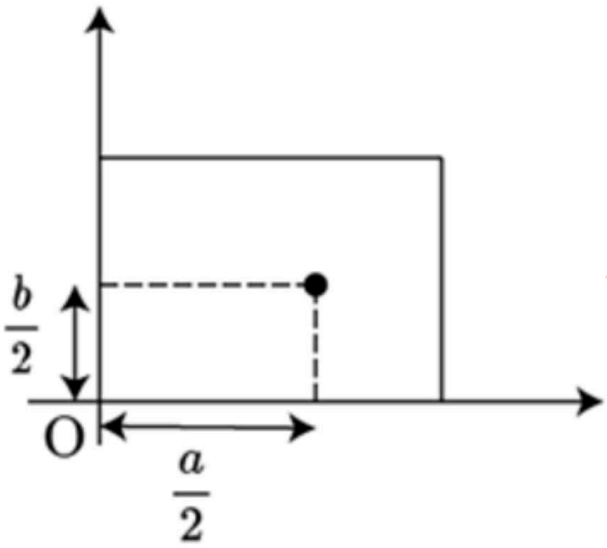
$\therefore g_E = g_h$

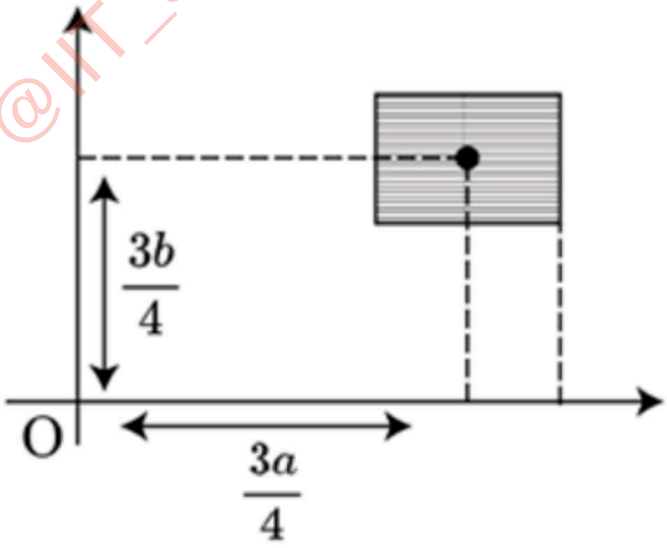
$$\therefore g - R\omega^2 = \left(1 - \frac{2h}{R}\right)g = g - \frac{2hg}{R}$$

$$\therefore R\omega^2 = \frac{2hg}{R} \therefore h = \frac{R^2\omega^2}{2g}$$

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Solution: If the total mass of object mass ' M ' then the mass removed part will be $\frac{M}{4}$.



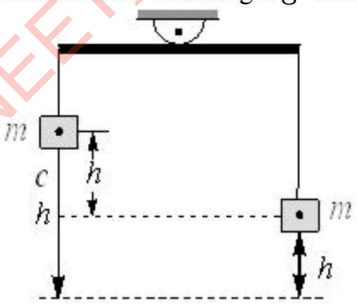


$$X_{CM}=\frac{M\cdot\frac{a}{2}-\frac{M}{4}\left(\frac{3a}{4}\right)}{M-\frac{M}{4}}=\frac{\frac{Ma}{2}-\frac{3Mb}{16}}{\frac{3M}{4}}=\frac{5}{12}\,a$$

$$Y_{CM}=\frac{M\cdot\frac{b}{2}-\frac{M}{4}\left(\frac{3b}{4}\right)}{M-\frac{M}{4}}=\frac{\frac{Mb}{2}-\frac{3Ma}{10}}{\frac{3M}{4}}=\frac{5}{12}\,b\,!$$

16. **Ans: (b)** $\frac{8}{3}\pi\rho Gmh$

Solution: As with height g varies as



and in accord with the figure $h_1 > h_2$ so W_1 will be lesser than W_2 and

$$W_2 - W_1 = mg_2 - mg_1 = 2mg\left[\frac{h_1}{R} - \frac{h_2}{R}\right]$$

or $W_2 - W_1 = 2m\,\frac{GM}{R^2}\,\frac{h}{R}$

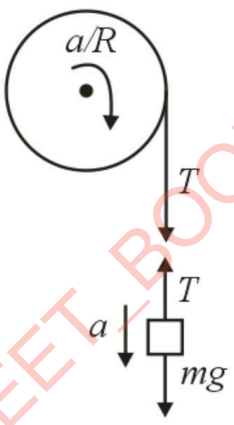
$$\left[\text{as } g = \frac{GM}{R^2} \text{ and } (h_1 - h_2) = h\right]$$

or $W_2 - W_1 = \frac{2mhG}{R^3}\left(\frac{4}{3}\pi R^3\rho\right)$

$= \frac{8}{3}\,\pi\rho Gmh\left[\text{as } M = \frac{4}{3}\pi R^3\rho\right]$ ^

17. **Ans: (a)** $\frac{2g}{3}$

Solution:



$mg - T = ma$ .....(1)

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$T \cdot R = 1/2\ mR^2\ \frac{a}{R}$ ....(2)

$\frac{2g}{3} = a$

/

18. **Ans: (d)** If both assertion and reason are false.

Solution: First, let ius look at them as two independent statements. During collision, there is deformity of shape of balls leading to increase in potential energy of the body, thus decreasing the kinetic energy momentarily. (Total energy remains constant, not KE) Thus, assertion is false. Energy spent against friction should be considered when calculating total energy. Law of conservation of energy always applies, in every case. Only note is to identify the forms of energy. Reason is also false. !

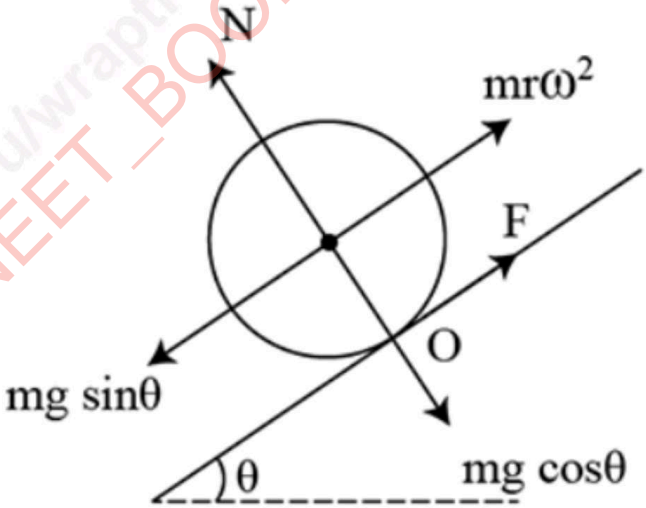
19. **Ans: (a)** $2.33\ \text{m s}^{-2}$

Solution: Free body diagram of the cylinder w.r.t. the inclined plane. The torque about the point of contact 0.

$(mg \sin\theta - mr\omega^2)R = I \cdot \alpha = \frac{3}{2}\ mR^2 \times \frac{a}{R} \quad (r = \text{radius of circle}, R = 6\ \text{m})$

$\therefore mg\sin\theta - mr\omega^2 = \frac{3}{2}\ ma$

$\therefore a = \frac{2}{3}\left(g \sin\theta - r\omega^2\right) = \frac{2}{3}\left[10 \times \frac{1}{2} - 6 \times \frac{1}{4}\right] = 2.33\ \text{m s}^{-2}$

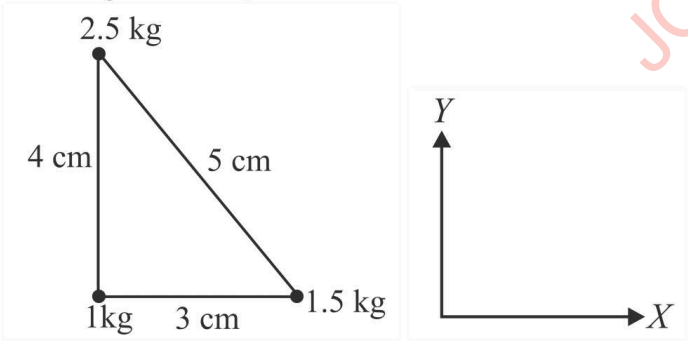


;

20. **Ans: (d)** $0.9\ \text{cm}$ right and $2.0\ \text{cm}$ above $1\ \text{kg}$ mass.

Solution:

Take $1\ \text{kg}$ mass at origin



$X_{cm} = \frac{1\times0+1.5\times3+2.5\times0}{5} = 0.9\ \text{cm}$

$Y_{cm} = \frac{1\times0+1.5\times0+2.5\times4}{5} = 2\ \text{cm}$

*

21. **Ans: 16**

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Solution: The energy corresponding to escape speed is

$$E = \frac{1}{2}\ m(2gR) = \frac{1}{2}\ m\left(2\frac{GM}{R^2}R\right)$$

$$= \frac{1}{2}\ m\left(2\frac{GM}{R}\right) = \frac{GMm}{R}$$

$$V_e = 64\ V_m$$

$$\Rightarrow \frac{4}{3}\pi R_e^3 = \frac{4}{3} \times \pi \times R_m^3 \times 64$$

$$\Rightarrow R_c = 4R_m$$

$$\because \rho_c = \rho_m$$

$$\frac{M_e}{V_e} = \frac{M_m}{V_m}$$

$$\Rightarrow \frac{M_e}{64\ V_m} = \frac{M_m}{V_m}$$

$$\Rightarrow M_c = 64M_m$$

$$\text{So } \frac{E_m}{E_e} = \frac{M_m}{R_m} \times \frac{R_e}{M_e}$$

$$= \frac{1}{64} \times 4 = \frac{1}{16} \quad \cdots\cdots [\because R_e = 4R_m]$$

$$E_m = \frac{E}{16}$$

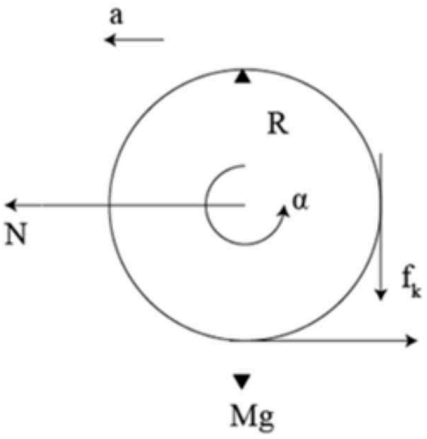
22. **Ans: 4**

Solution: For translation, $N - f_s = ma$ for rotation $(f_s - f_k)R = I\alpha$ and $a = R\alpha$.
Solving, $f_k = 0.8$ N; $f_s = 1.4$ N

$$\Rightarrow \mu = \frac{P}{10} = \frac{f_k}{2\ N}$$

$$\Rightarrow \frac{P}{10} = \frac{0.8}{2}$$

$$\Rightarrow P = 4$$



23. **Ans: 6**

Solution:

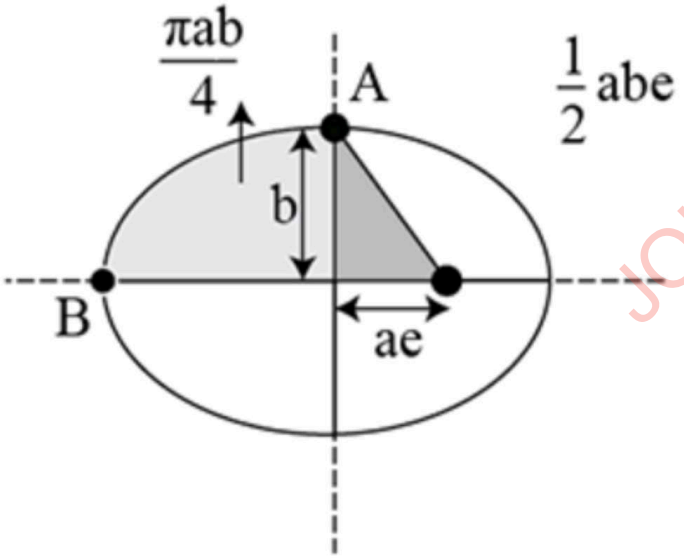
$$t = \frac{T}{\pi ab}\left(\frac{\pi ab}{4} + \frac{1}{2}abe\right)$$

$$t = T\left(\frac{1}{4} + \frac{e}{2\pi}\right)$$

$$t = 16\left(\frac{1}{4} + \frac{1}{8}\right)$$

$$t = 6 \text{ months}$$

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24. **Ans: 50**

Solution:

$$I_P = \left[\frac{4mR^2}{2} + m\left(4R^2\right)\right] - \left[\frac{m}{4}\frac{R^2}{2} + \frac{m}{4}5R^2\right]$$

$$I_P = mR^2\left[(2+4) - \left(\frac{1}{8} + \frac{5}{4}\right)\right]$$

$$I_P = mR^2\left(6 - \frac{11}{8}\right) = \frac{37}{8}mR^2 \qquad \ldots(1)$$

$$I_O = \left(\frac{4mR^2}{2}\right) - \left(\frac{m}{4}\frac{R^2}{2} + \frac{m}{4}R^2\right)$$

$$I_O = mR^2\left[2 - \left(\frac{1}{8} + \frac{1}{4}\right)\right] = mR^2\left[2 - \frac{3}{8}\right] = mR^2\left(\frac{13}{8}\right) \quad \ldots(2)$$

$$\text{So } \frac{I_P}{I_O} = \frac{37/8}{13/8} = \frac{37}{13}$$

$$\Rightarrow m + n = 37 + 13 = 50$$

25. **Ans: 35**

Solution: Applying conservation of linear momentum at the time of the collision, or at $t = 1\text{ s}$, $m\vec{v} + m(0) = 2\,m(20\hat{i} + 10\hat{j})$

$\therefore \quad \vec{v} = 40\hat{i} + 20\hat{j}$

At 1sec, masses will be at height : $h_1 = u_y t + \frac{1}{2}a_y t^2 = (20)(1) + \frac{1}{2}(-10)(1)^2 = 15\text{ m}$

After explosion other mass will further rise to a height: $h_2 = \frac{u_y^2}{2g} = \frac{(20)^2}{2\times 10} = 20\text{ m}$:

$u_y = 20\text{ m/s}$ just after the collision.

$\therefore$ Total height $h = h_1 + h_2 = 35\text{ m}$

26. **Ans: 4**

Solution:

Loss of KE = Gain in PE

$$\Rightarrow \frac{1}{2}mv^2 = mgh\left(\frac{R}{h+R}\right)$$

CASE I.

$$\Rightarrow \frac{1}{2}m\left(\frac{2gR}{3}\right) = mgh_1\left(\frac{R}{h_1+R}\right) \quad \ldots(i)$$

$$\Rightarrow \frac{1}{2}m\left(\frac{4gR}{3}\right) = mgh_2\left(\frac{R}{h_2+R}\right) \quad \ldots(ii)$$

Solving (i)

$$\Rightarrow \frac{1}{3} = \frac{h_1}{h_1+R}$$

$$\Rightarrow h_1 = \frac{R}{2} \quad \ldots(iii)$$

Solving (ii)

$\Rightarrow \frac{2}{3} = \frac{h_2}{h_2+R}$ ...(iv)

$\Rightarrow h_2 = 2R$

From (iii) and (iv)

$\Rightarrow \frac{h_1}{h_2} = \frac{R/2}{2R}$

$\frac{h_2}{h_1} = 4$

27. **Ans: 32**

Solution:

$$\vec{L} = I_{CM} \cdot \omega + mv_{CM}R$$

$$\omega = \frac{v}{R} = \frac{2}{4} = 0.5 \text{rads}^{-1}$$

$$\vec{L} = mR^2 \cdot \omega + mv_{CM} \cdot R$$

$$= 2 \times 16 \times 0.5 + 2 \times 2 \times 4 = 16 + 16$$

$$\vec{L} = 32 \text{ kg m}^2 \text{ s}^{-1}$$

28. **Ans: 24**

Solution: Momentum of one bullet, $p = mv$

$$= 20 \times 10^{-3} \times 300 = 6 \text{ kg m s}^{-1}$$

$N =$ number of bullet per sec $= 4$

$\therefore \frac{dp}{dt} =$ change of momentum per sec or force

$= N(p - 0) = 4 \times 6 = 24 \text{ N}$

29. **Ans: 4**

Solution: Acceleration due to gravity is $g = \frac{GM}{R^2}$

and if L be the angular momentum of the earth, then rotational $\text{KE} = \frac{L^2}{2I'}$ where, I be the moment of inertia.

So, $I = \frac{2}{5} MR^2$ (for sphere) $I = \frac{2}{5} MR^2$

$\therefore$ Rotational $\text{KE} = \frac{5L^2}{4MR^2}$

Since, angular momentum remains conserved.

So, rotational $\text{KE}(K) \propto \frac{1}{R^2}$

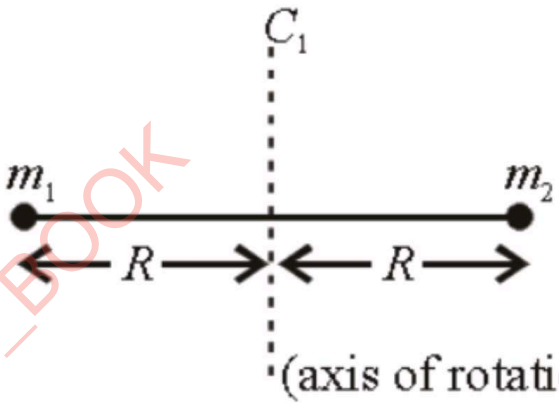
$\therefore$ Both g and K are $\propto R^{-2}$.

$\therefore \frac{\Delta g}{g} = \frac{\Delta K}{K} = -2 \times \frac{\Delta R}{R}$

$\therefore$ Both g and K would decrease by

$2 \times 2\% = 4\%$

30. **Ans: 4**



Solution:

Suppose the masses m_1 and m_2 are attached to the ends of a massless rod. C_1 is the axis of rotation which is at a distance R from each of the masses. $\therefore$ the moment of inertia of the system, $\text{I} = m_1R^2 + m_2R^2 = (m_1 + m_2)R^2$ Now if the distance between the

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masses is doubled, then the moment of inertia $\Gamma = m_1(2R)^2 + m_2(2R)^2 = (m_1 + m_2)4R^2$ Since the same torque is applied in both the cases, $\tau = I\alpha = I'\alpha'$

$$\Rightarrow \alpha' = \frac{I\alpha}{I'} = \frac{(m_1 + m_2)R^2\alpha}{(m_1 + m_2)4R^2} = \frac{\alpha}{4}$$

31. **Ans: (a)** 0.02

Solution: $K_{sp_1} = \left[Ba^{2+}\right]\left[SO_4^{2-}\right]$

$K_{sp_2} = \left[Ba^{2+}\right]\left[CO_3^{2-}\right]$

$\frac{K_{sp_1}}{K_{sp_2}} = \frac{SO_4^{2-}}{CO_3^{2-}} = \frac{1\times10^{-10}}{5\times10^{-9}} = 0.02$

32. **Ans: (c)** The volume on the reaction mixture was decreased

Solution: Volume of the vessel was decreased, which led to sudden increase in $[H_2]$. The reaction then moves in direction to achieve a state of new equilibrium.
A catalyst does not affect the equilibrium concentration of reactants and products.
Increase in pressure would decrease concentration as reaction favoured forward.
$P_4(s)$ will have no effect on k_p and concentration of $H_2(g)$ (because active mass of solid is taken to be unit)

33. **Ans: (d)** -48

$$N_2O_{4(g)} + 3CO_{(g)} \longrightarrow N_2O_{(g)} + 3CO_{2(g)}$$

Solution: $\Delta H_{\text{reaction}} = \Sigma_{\text{Heat of formation of products}} - \Sigma_{\text{Heat of formation of reactants}}$

$$\Delta H_{\text{reaction}} = \left[\Delta H_f\, N_2O + 3 \times \Delta H_f CO_2\right] - \left[\Delta H_f\, N_2O_4 + 3 \times \Delta H_f CO\right]$$

$$\Delta H_r = [+811 + 3(-393)] - [10 + 3(-110)]$$
$$= [811 - 1179] - [-320] = -368 + 320$$
$$= -48\ \text{kJ/mol}$$

34. **Ans: (a)** 12

Solution: $40\ \text{mL}$ of $0.1\text{M}HCl = \frac{0.1}{1000} \times 40$
$= 0.004$ mole
$10\ \text{mL}$ of $0.45\text{M}NaOH = \frac{0.45}{1000} \times 10 = 0.0045$ mole
$\therefore NaOH$ left unused $= 0.0005$ mole
Volume of solution $= 50\ \text{mL}$
$\therefore$ Molarity of $NaOH = \frac{0.0005}{50} \times 1000$
$= 0.01\text{M}$
Or $[OH^-] = 10^{-2}$ or $[H^+] = 10^{-12}$ $\quad pH = 12$

35. **Ans: (a)** $a \to r, s; b \to p, s; c \to r, s; d \to q$

Solution:

$(a \to r)$

Enthalpy is a state function and this is a function of internal energy, pressure and volume.

$(b \to p)$

Temperature is independent of bulk matter of the system.

Hence, it is an intrinsic property but it is a state variable.

$(c \to r)$

Free energy is a state function as it depends on the state of the system as well as H,T and S.

It is also a function of U, P and V.

$(d \to q)$

Work is a path function as the amount of work done depends upon the path undertaken.

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36. **Ans: (b)** 11.3

Solution: $\mathrm{pH}=2,[\mathrm{HCl}]=10^{-2}(\mathrm{M})$ & $\mathrm{pOH}=2$
or $[\mathrm{NaOH}]=10^{-2}(\mathrm{M})$
4ml of $10^{-2}(\mathrm{M})\mathrm{HCl} \equiv 4\times10^{-5}$ moles HCl.
6ml of $10^{-2}(\mathrm{M})\mathrm{NaOH} \equiv 6\times10^{-5}$ moles NaOH
After mixing total moles of OH^- = moles of NaOH − moles of $\mathrm{HCl}=6\times10^{-5}-4\times10^{-5}$
After mixing excess moles of $\mathrm{OH}^-=2\times10^{-5}$ Total volume $=6+4=10\mathrm{ml}=10\times10^{-3}$ litre
$[\mathrm{OH}^-]=\frac{2\times10^{-5}}{10}\times10^3=2\times10^{-3}$
or $\quad \mathrm{pOH}=3-\log 2=3-0.3=2.7$
or $\mathrm{pH}=11.3$

37. **Ans: (c)** 0.7 atm

Solution: $2\mathrm{AB(s)} \rightleftharpoons \mathrm{A_2(\,g)}+\mathrm{B_2(\,g)}$
$(0.5+p_0)\mathrm{p_0}$
$K_p=p_{A_2}\times p_{B_2}$
$0.06=(0.5+p_0)p_0$
$p_0^2+0.5p_0-0.06=0$
$p_0=0.1$
$p_{\text{total}}=p_{A_2}+p_{B_2}=(0.5+p_0)+p_0$
$=0.6+0.1=0.7$ atm

38. **Ans: (b)** $-74.8\ \mathrm{kJ\ mol^{-1}}$

Solution: For reaction
$\mathrm{CO_2(g)+2H_2O(l)\rightarrow CH_4(g)+2O_2(g)}$
$\Delta_r\mathrm{H^o}=\Sigma(\Delta_f\mathrm{H^o})_{\text{products}}-\Sigma(\Delta_f\mathrm{H^o})_{\text{reactants}}$
$=[(\Delta_r\mathrm{H^o(CH_4)}+2\times0)-(\Delta_f\mathrm{H^o(CO_2)}+2\Delta_f\mathrm{H^o(H_2O)})]$
$+890.3=[\Delta_f\mathrm{H^o(CH_4)}]-[-393.5+2\times(-285.8)]$
$\Delta_f\mathrm{H^o}$ of $\mathrm{CH_4}$ (g) $=-74.8\ \mathrm{kJ/mol}$

39. **Ans: (a)** $8.5\times10^{-5}\mathrm{M}$

Solution: $K_{\text{sp}}(\mathrm{SrCrO_4})=\left[\mathrm{Sr^{2+}}\right]\left[\mathrm{CrO_4^{2-}}\right]$
$\left[\mathrm{CrO_4^{2-}}\right]=\frac{7\times10^{-6}}{0.1}=7\times10^{-5}$
$K_{\text{sp}}(\mathrm{BaCrO_4})=\left[\mathrm{Ba^{2+}}\right]\left[\mathrm{CrO_4^{2-}}\right]$
$\left[\mathrm{CrO_4^{2-}}\right]_{\text{total}}\approx\left[\mathrm{CrO_4^{2-}}\right]$ from $\mathrm{SrCrO_4}$
$\left[\mathrm{Ba^{2+}}\right]=\frac{K_{\text{sp}}(\mathrm{BaCrO_4})}{[\mathrm{CrO_4^{2-}}]}$
$\left[\mathrm{Ba^{2+}}\right]=\frac{6\times10^{-9}}{7\times0^{-5}}=0.85\times10^{-4}$
$=8.5\times10^{-5}\mathrm{M}$

40. **Ans: (a)** $\mathrm{N_2(\,g)+3H_2(\,g)\rightarrow 2NH_3(\,g)}$

Solution: Entropy is the measure of randomness. As the randomness (disorder) increases, entropy increases, The order of entropy is
$\underset{\text{prderticles}}{\text{Solid}}<\text{Liquid}<\underset{\text{(most disordered)}}{\text{Gas}}$
Thus, in reaction $\mathrm{N_2(\,g)+3H_2(\,g)\rightarrow 2NH_3(\,g)}$
Entropy decreases because number of gaseous molecules decreases.

41. **Ans: (a)** 0.120

Solution: This problem is based on the concept of degree of dissociation and equilibrium constant. It is advisable to follow the given steps to solve the question.
Write the decomposition reaction then calculaate total pressure at equilibrium in terms of degree of dissociation (α).
Then, calculate pressure of $\mathrm{NO_2}$ $(\mathrm{p_{NO_2}})$ using value of Kp and p.
Then after calculate degree of dissociation (α).
Given reaction is

	$\mathrm{N_2O}$	$\rightleftharpoons$	$2\mathrm{NO_2}$
At eqbuilibrium	$1-\alpha$		2α

Let α = degree of dissociation, p = total pressure
Total number of moles $=2\alpha+1-\alpha=1+\alpha$

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$$\therefore \quad p_{NO_2} = \left(\frac{1-\alpha}{1+\alpha}\right) \cdot p$$

$$= \frac{\left[p_{NO_2}\right]}{p_{NO_2}} = \frac{\left[\frac{(1-\alpha)}{(1+\alpha)}\, p\right]^2}{\frac{1-\alpha}{1+\alpha}} = \frac{4\alpha^2 p^2}{1+\alpha} \times \frac{1+\alpha}{(1-\alpha)\, p}$$

$$= \frac{4\alpha^2 p}{1-\alpha^2}$$

Putting values of Kp and p, we get

$$0.120 = \frac{4\alpha^2 (2)}{1-\alpha^2}$$

$$= \frac{8\alpha^2}{1-\alpha^2}$$

$$0.120\left(1-\alpha^2\right) = 8\alpha^2$$

$$\alpha = \left(\frac{0.120}{8.12}\right)^{1/2} = 0.120$$

Hence, 0.120 is the choice.

42. **Ans: (b)** Change in internal energy is same in all the four cases

Solution: As work done $=$ area under $p-V$ curve, work done is maximum in case I Change in internal energy is independent of the path from A to B. Therefore, in all cases change in internal energy is same.

43. **Ans: (d)** $1:10$

Solution: Given, that $pH = 5.74, pK_a = 4.74$
Suppose that volume of acid solution $= x L$ Volume of salt solution $= yL$
From Henderson equation,

$$pH = pK_a + \log \frac{[Salt]}{[Acid]}$$

or, $$pH - pK_a = \log \frac{[CH_3COONa]}{[CH_3COOH]}$$

or, $$5.74 - 4.74 = 1 = \log \frac{[CH_3COONa]}{[CH_3COOH]}$$

or $$\frac{CH_3COONa]}{[CH_3COOH]} = 10$$

or $$\frac{[CH_3COOH]}{[CH_3COONa]} = \frac{1}{10} = \frac{\frac{x+y}{0.1y}}{x+y}$$

Thus, $\frac{x}{y} = \frac{1}{10}$

44. **Ans: (d)** I, II and III are the same

Solution: As Internal energy change is a state function so it depends only on the initial and final temperature and does not depends on the path taken.

45. **Ans: (a)** ClO^-

Solution: $HClO_4$ is the strongest acid among $HClO$, $HClO_2$, $HClO_3$ and $HClO_4$ as more the number of oxygen atoms attached, more is the acidity of the molecule order of acidity is : $HClO < HClO_2 < HClO_3 < HClO_4$ So the conjugate base order will be just the reverse: $ClO^- > ClO_2^- > ClO_3^- > ClO_4^-$

46. **Ans: (d)** All

Solution: Here all the given statements are true. The entropy of isolated system with P-V work only, is always maximized at equilibrium. It is possible for the entropy of closed system to decrease substantialy in an irreversible process. By second law of thermodynamics it can be said that entropy can be created but not destroyed. ΔS_{system} is zero for reversible process in an isolated system.

47. **Ans: (d)** If both Assertion and Reason are false statements, then mark (4).

Solution: Value of equilibrium constant does not depend on concentration of reactants or products.

48. **Ans: (d)** $0.05M\ KClO_4 + 0.05M\ HClO_4$

Solution: Buffer solutions can be obtained by mixing a weak acid with its salt formed with a strong base or by mixing a weak base with its salt formed with a strong acid. As $HClO_4$ is a strong acid, therefore equimolar mixture of $HClO_4$ and its salt $KClO_4$ is not a buffer solution.

49. **Ans: (d)** $750\ K$

Solution: $\frac{1}{2} X_2 + \frac{3}{2} Y_2 \longrightarrow XY_3$

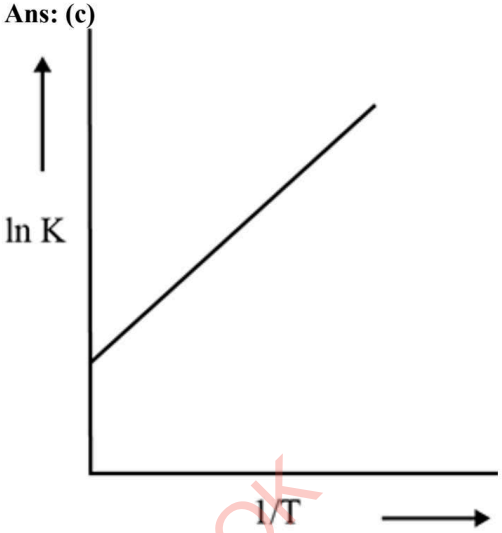
$$\Delta S^\circ_{\text{reaction}} = \Delta S^\circ_{\text{products}} - \Delta S^\circ_{\text{reactants}}$$

$$\therefore \Delta S^\circ_{\text{reaction}} = \Delta S^\circ_{XY_3} - \frac{1}{2}\Delta S^\circ_{X_2} - \frac{3}{2}\Delta S^\circ_{Y_2}$$

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$= 50 - \frac{1}{2} \times 60 - \frac{3}{2} \times 40 = -40\ \mathrm{JK^{-1}}$
Using equation, $\Delta G = \Delta H - T\Delta S$
We have, $\Delta H = -30\ \mathrm{kJ}$, $\Delta S = -40\ \mathrm{JK^{-1}\ mol^{-1}}$
and at equilibrium, $\Delta G = 0$. Therefore,
$T = \frac{\Delta H}{\Delta S} = \frac{-30 \times 1000}{-40} = 750\ \mathrm{K}$

50. **Ans: (c)**



Solution: $\mathrm{InK} = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T}\right) + \frac{\Delta S^\circ}{R}$
At high temperature if shifted backward, it is exothermic reaction.
ΔH°_{reac} is -ve
Slope is +ve

51. **Ans: 8**

Solution: $K_a(C_6H_5COOH) = 1 \times 10^{-4}$
pH of $0.01M\ C_6H_5\ COONa$
$K_h = \frac{K_w}{K_a} = \frac{0.01\ h^2}{1-h}$
$\frac{10^{-14}}{10^{-4}} = \frac{10^{-2}h^2}{1-h}\ (1 - h \quad 1)$
$[OH^-] = 0.01\ h = 0.01 \times 10^{-4} = 10^{-6}$
$[H^+] = 10^{-8} pH = 8$

52. **Ans: 2**

Solution: $K_p = K_c(RT)^{\Delta n}$
$\Delta n = \left\{ \begin{array}{c} \text{No. of gaseous} \\ \text{moles of products} \end{array} \right\} - \left\{ \begin{array}{c} \text{No. of gaseous} \\ \text{moles of reactants} \end{array} \right\}$
$= x$
$K_p = K_c(RT)^x = K_c(RT)^2$
$\therefore\ x = 2$

53. **Ans: 1250**

Solution: $C_p = q/T;$
Heat required = No. of moles $\times C_p \times T$
$= (144 \times 25 \times 10)/(14.4 \times 2) = 1250\ \mathrm{J}$

54. **Ans: 4**

Solution: $pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$
$= -\log 10^{-10} + \log 1 = 10; pH = 14 - 10 = 4$

55. **Ans: 60**

Solution: Number of moles of $CaCO_3$ present initially $= \frac{50}{100} = 0.5$ moles K_p of the given reaction can be written as:
$K_p = P_{CO_2} = 1.64$ atm Number of moles of $CO_2 = \frac{P_{CO_2}\ V}{RT} = \frac{1.64 \times 10}{0.082 \times 1000} = 0.2$ Number of moles of $CaCO_3$ reacted $= 0.2$ moles Number of moles of $CaCO_3$ unreacted $= (0.5 - 0.2) = 0.3$ moles Percentage of $CaCO_3$ unreacted $= \frac{0.3}{0.5} \times 100 = 60\%$

56. **Ans: 8**

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$Q = 0 \quad (\because$ process is adiabatic. $)$

Solution: $\Delta U = w = -P_{\text{ext}}\Delta V$

$= -1(10-2)$ atm L $= -8$ atm L

57. **Ans: 230**

Solution:

Given reaction:

$3CaO + Al \rightarrow Al_2O_3 + 3Ca$

Now, $\Delta_r H^\circ = \Sigma\Delta_f H^\circ_{\text{Products}} - \Sigma\Delta_f H^\circ_{\text{Reactants}}$

$= [1 \times (-1675) + 3 \times 0] - [3 \times (-635) + 2 \times 0]$

$= +230 \text{ kJ mol}^{-1}$

58. **Ans: 9**

Solution: $0.1 \times 1 = (1 + \text{v}) \times 0.01 \Rightarrow 1 + \text{v} = \frac{0.1}{0.01}$
$\Rightarrow 1 + v = 10 \Rightarrow v = 10 - 1 = 9L$

59. **Ans: 37**

Solution:

Amount of CO_2 in one liter of solution $= 4.4. = 0.1$ Mole

$\text{pH} = 1/2\{\text{pKa} - \log C\}$

For a weak acid solution

$\text{pH} = 1/2\{6.4 + 1\} = 3.7$

60. **Ans: 815**

Solution: $2C(s) + H_2(g) \longrightarrow C_2H_2(g)(H - C \equiv C - H)$
$\Rightarrow B \cdot E(H_2) + \Delta B \cdot E(H_2) + \Delta H_{\text{sub}}(C) - B \cdot E(C - H) \times 2 + B \cdot E(C \equiv C)\Delta H_{\text{rxn}}$
$\Rightarrow 330 + 1410 - [350 \times 2 + x] = 225 \Rightarrow x = 815$

61. **Ans: (b)** $\frac{9!}{2!\,4!}$

Solution:

C	I	N	O	R	S	U
1	2	2	1	1	1	1

Total number of arrangements $= \frac{9!}{2!2!}$
The number of ways, vowels in these words are arranged
$= \frac{4!}{2!}$

$\begin{pmatrix} I & O & U \\ 2 & 1 & 1 \end{pmatrix}$

But the vowels are restricted to have only one arrangement i.e. IIOU
$\Rightarrow$ Total number of possible arrangement of the letters of the word INCURSION $= \frac{9!/2!2!}{4!/2!} = \frac{9!}{2!4!}$

62. **Ans: (d)** 1140

Solution:

The number of solutions (positive integer) of $x + y + z = r$ is ${}^{r-1}C_2$
$\therefore$ The desired number of solutions is
$\sum_{r=3}^{20} {}^{r-1}C_2$

${}^2C_2 + {}^3C_2 + {}^4C_2 + ... + {}^{19}C_2$

$= \left({}^3C_3 + {}^3C_2\right) + {}^4C_2 + ... + {}^{19}C_2$

$= {}^4C_3 + {}^4C_2 + ... + {}^{18}C_2 + {}^{19}C_2$

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$= {}^{20}C_3$ (Adding in the same way)

$= \frac{20\times19\times\cancel{18}3}{\cancel{3}\times\cancel{2}} = 1140$

63. **Ans: (c)** $(\cot 2, 0)$

Solution:

$\cos^{-1}x > \cos^{-1}x^2 \Rightarrow x < x^2 \;\&\; x \in [-1,1]$

$\{\because \cos^{-1}x$ is a decreasing function$\}$

$\Rightarrow x(x-1) > 0 \;\&\; x \in [-1,1]$

$\Rightarrow x \in (-\infty, 0) \cup [1, \infty) \;\&\; x \in [-1, 1]$

$\Rightarrow x \in [-1, 0) \ldots . (1)$

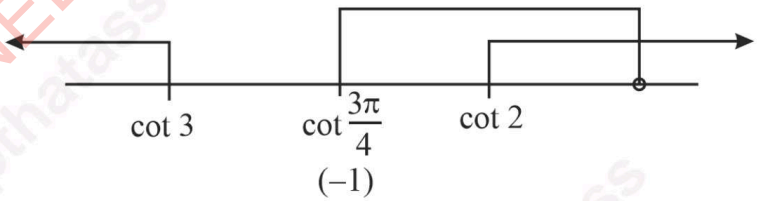
Now $\left(\cot^{-1}x\right)^2 - 5\cot^{-1}x + 6 > 0$

$\Rightarrow \left(\cot^{-1}x - 2\right)\left(\cot^{-1}x - 3\right) > 0$

$\Rightarrow \cot^{-1}x < 2$ or $\cot^{-1}x > 3$

$x > \cot 2$ or $x < \cot 3 \ldots (2)$

$\{\because \cot^{-1}x$ is a decreasing function$\}$



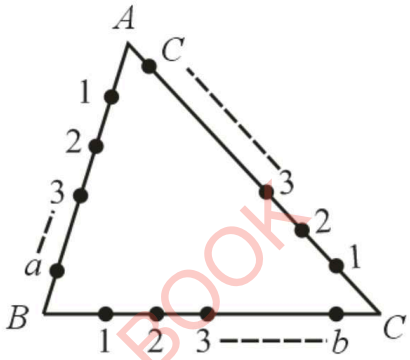
$(1) \cap (2) \Rightarrow x \in (\cot 2, 0)$

64. **Ans: (b)** $\frac{29}{180}$

Solution: We need 3 -digit number which is divisible by $4\&5$ both. i.e. their last two digits are $00, 20, 40, 60\&80$ Now, ending with 00 are '9'. $\{100, 200, \ldots\ldots, 900\}$ If digit repeat other than ' 0 ' then they are $\{220, 440, 660, 880\}$ but 220 numbers can be permuted according to the condition as $\{220, 202\}$ So, there are ' 8 ' other favorable cases. If the number have no digit repeated like 320 . 320 can be permuted in 4 ways. $\{302, 230, 320, 203\}$ So, such numbers are $8 \times 4 \times 4 = 128$ Total favorable $= 9 + 8 + 128 = 145$ So, required prob. $= \frac{145}{900} = \frac{29}{180}$

65. **Ans: (d)** None of These

Solution:



Required number of triangles

=Total number of ways of choosing 3 points−Number of ways of choosing all the 3 Points from one of AB or BC or CA

$= {}^{a+b+c}C_3 - \left({}^aC_3 + {}^bC_3 + {}^cC_3\right)$

66. **Ans: (a)** $\cot^{-1}1$

Solution:

Let $T_r = \sum_{r=1}^{\infty} \cot^{-1}\left(3r^2 - r - \frac{1}{3}\right) = \tan^{-1}\left(\frac{3}{9r^2-3r-1}\right)$

$= \tan^{-1}\left(\frac{(3r+1)-(3r-2)}{1+(3r+1)(3r-2)}\right)$

$\sum_{r=1}^{\infty} T_r = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} = \tan^{-1}1 = \cot^{-1}1$

67. **Ans: (d)** 91

Solution: $P(\overline{EE}) + P(\overline{EEEEE})$

$= \frac{5}{6} \times \frac{1}{6} + \left(\frac{5}{6}\right)^4 \times \frac{1}{6} + \left(\frac{5}{6}\right)^8 \frac{1}{6} \ldots \infty$

$= \frac{5}{36}\left[1 + \left(\frac{5}{6}\right)^3 \ldots\right] = \frac{30}{91} = 0.329 = 0.33$

68. **Ans: (a)** 6

Solution:

Green$\rightarrow$ g, Blue $\rightarrow 3$

$\frac{1}{2} = \frac{{}^{3}C_1 \cdot {}^{g}C_1}{(g+3)_{C_2}}$

$\frac{(g+3)(g+2)}{2} = 6g$

$9^2 + 5g + 6 = 12g \Rightarrow g^2 - 7g + 6 = 0$

$g = 1$ or 6

69. **Ans: (b)** 600

Solution: $a + c = 2b$
So, both a and c will be even or both will be odd as their sum is even.
In first 50 natural number 25 are even and 25 are odd. Required number of ways ${}^{2}C_{52} + {}^{2}C_{52} = 2 \times \frac{25\times24}{2} = 600$

70. **Ans: (a)** $\frac{6}{55}$

Solution: Total cases $= 11!$
Number of favourable cases $= 2! \times {}^{9}C_4 \times 4! \times 6!$
Hence, required probability $= \frac{2!\times{}^{9}C_4\times4!\times6!}{11!}$
$= \frac{2!\times9!\times4!\times6!}{11!\times5!\times4!}$
$= \frac{2\times6}{11\times10} = \frac{6}{55}$

71. **Ans: (a)** 133320

Solution: There are $4! = 24$ numbers. Each digit occurring $3! = 6$ times, in the unit's, ten's, hundred's and thousand's places. We note that $(2 + 4 + 6 + 8) = 120$. Thus in the over all sum there will be 120 units, 120 tens, 120 hundreds and 120 thousands.
The required sum
$= 120\left(1 + 10 + 10^2 + 10^3\right)$
$= 120 \times 1111 = 133320$

72. **Ans: (a)** A and B are independent

Solution: Given, $P(A \cap B) = \frac{1}{6}$
$P(A \cup B) = \frac{31}{45}$
and $P(\bar{B}) = \frac{7}{10}$
$\therefore \quad P(B) = 1 - \frac{7}{10} = \frac{3}{10}$
Now, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$\Rightarrow \quad \frac{31}{45} = P(A) + \frac{3}{10} - \frac{1}{6}$

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$\therefore \quad P(A) = \frac{31}{45} + \frac{1}{6} - \frac{3}{10} = \frac{5}{9}$

Then. $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{1}{5} = \frac{3}{10} > \frac{1}{6}$

and $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{6}}{\frac{3}{3}} = \frac{5}{9} > \frac{1}{6}$

and, $P(A) \times P(B) = \frac{5}{9} \times \frac{3}{10} = \frac{1}{6} = P(A \cap B)$

Hence, A and B are independent.

73. **Ans: (d)** 256

Solution: Given, $x_1 . x_2 . x_3 . x_4 = 770 = 2.5.7.11$

Let positive integer $x_1 = 2^{a_1}.5^{b_1}.7^{c_1}.11^{d_1}$

Let positive integer $x_2 = 2^{a_2}.5^{b_2}.7^{c_2}.11^{d_2}$

Let positive integer $x_3 = 2^{a_3}.5^{b_3}.7^{c_3}.11^{d_3}$

Let positive integer $x_4 = 2^{a_4}.5^{b_4}.7^{c_4}.11^{d_4}$

Now, $x_1 x_2 x_3 x_4 = 2^{a_1+a_2+a_3+a_4}.5^{b_1+b_2+b_3+b_4}.7^{c_1+c_2+c_3+c_4}.11^{d_1+d_2+d_3+d_4}$

As per given condition $a_1 + a_2 + a_3 + a_4 = 1$, which can be in 4 ways $(1,0,0,0), (0,1,0,0)(0,0,1,0)$ and $(0,0,0,1)$, similarly for other powers.

Hence total number of ways $= 4 \times 4 \times 4 \times 4 = 256$

74. **Ans: (b)** $5 - 2\pi$

Solution:

$$\tan^{-1}\left(\tan\left(f(-5) + f(20) + \cos^{-1}(f(-10) + f(17))\right)\right)$$

$$= \tan^{-1}\left(\tan\left(2 + 3 + \cos^{-1}(1-2)\right)\right) = \tan^{-1}(\tan(5+\pi)) = \tan^{-1}(\tan 5) = 5 - 2\pi$$

75. **Ans: (b)** $\frac{1}{14}$

Solution:

Total ways of arrangements $= \frac{8!}{2!.4!}$.

w, x, y, z

Now 'S' can have places at dot's and in places of $w,\ x,\ y,\ z$ we have to put $2A's$, one I and one N

Therefore, favourable ways $= {}^5C_4 \times \left(\frac{4!}{2!}\right)$

Hence, required probability $= \frac{5 \cdot 4! \cdot 2! \cdot 4!}{2! \cdot 8!} = \frac{1}{14}$.

76. **Ans: (c)** 13

Solution: $N = ababab\ \ 1 < a \leq 9 \quad 0 < b \leq 9 \quad a, b \in I\ N = 10^5 a + 10^4 b + 10^3 a + 10^2 b + 10a + b$

$= \left(10^4 + 10^2 + 1\right)(10a + b) = \left(10^2 + 10 + 1\right)\left(10^2 - 10 + 1\right)(10a + b) = 3 \times 37 \times 13 \times 7(10a + b) \quad \ldots\ldots (1)$

then $\quad 10\mathrm{a} + \mathrm{b} = \mathrm{P}_1 \times \mathrm{P}_2 \quad \mathrm{p}_1, \mathrm{p}_2 \in$ prime and $10 \leq 10\mathrm{a} + \mathrm{b} \leq 99$

a	b	$10\mathrm{a} + \mathrm{b}$
1	0	$10 = 2 \times 5$
2	2	$22 = 2 \times 11$
3	4	$34 = 2 \times 17$
3	8	$38 = 2 \times 19$
4	6	$46 = 2 \times 33$
5	5	$55 = 5 \times 11$
5	8	$58 = 2 \times 29$
6	2	$62 = 2 \times 31$
7	4	$74 = 2 \times 37$
8	2	$82 = 2 \times 41$
8	5	$85 = 5 \times 17$
9	4	$94 = 2 \times 47$
9	5	$95 = 5 \times 19$

77. **Ans: (b)** $\frac{8}{17}$

Solution:

$P(B_1) = \frac{1}{2} = P(B_2)$

Box B_1 have 20 non-prime numbers $\{1, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30\}$

$\Rightarrow P(NP/B_1) = \frac{20}{30}$

Box B_1 have 15 non-prime numbers $\{32, 33, 34, 35, 36, 38, 39, 40, 42, 44, 45, 46, 48, 49, 50\}$

$\Rightarrow P(NP/B_2) = \frac{15}{20}$

Probability of non-prime$= P(NP)$.

$P(NP) = P(B_1).P(NP/B_1) + P(B_2).P(NP/B_2)$

$= \frac{1}{2} \cdot \frac{20}{30} + \frac{1}{2} \cdot \frac{15}{20}$

$P(B_1/N.P) = \frac{\frac{1}{2} \cdot \frac{20}{30}}{\frac{1}{2} \cdot \frac{20}{30} + \frac{1}{2} \cdot \frac{15}{20}} = \frac{8}{17}$

78. **Ans: (a)** 23

Solution:

$1008 = 2^4 \times 3^2 \times 7$

$\therefore$ The required number of even proper divisors
$=$ Total number of selections of atleast one 2 and any number of 3's or 7's

$= 4 \times (2+1) \times (1+1) - 1 = 23.$

79. **Ans: (d)** $15 + \frac{25}{24}$

Solution:

$$\text{Given expression } = 1 + 2^2 + 1 + 3^2 + \operatorname{cosec}\left(\tan^{-1}\frac{4}{3} + \tan^{-1}\frac{4}{3}\right)$$

$$= 15 + \operatorname{cosec}\left(\pi + \tan^{-1}\left(\frac{-24}{7}\right)\right) = 15 + \operatorname{cosec}\left(\operatorname{cosec}^{-1}\frac{25}{24}\right) = 15 + \frac{25}{24}$$

80. **Ans: (c)** 14

Solution:

k	2	3	4	5	6	7	8	9	10	11	12
$P(k)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$k =$ sum on dice
Expected value $= \frac{1}{36} \times 4 + \frac{2}{36} \times 6 + \frac{3}{36} \times 8 + \frac{4}{36} \times 10$
$+ \frac{5}{36} \times 12 + \frac{6}{36} \times 14 + \frac{5}{36} \times 16 + \frac{4}{36} \times 18$
$+ \frac{3}{36} \times 20 + \frac{2}{36} \times 22 + \frac{1}{36} \times 24$
$= \frac{504}{36} = 14$

81. **Ans: 28**

Solution: Let the probability of occurrence of first event A, be 'a'
i.e., $P(A) = a$
$\therefore P(\text{not } A) = 1 - a$
And also suppose that probability of occurrence of second event B, $P(B) = b$,
$\therefore$ P(not B) $= 1 -$ b
Now, $P(A$ and not $B) + P($ not A and $B) = \frac{26}{49}$

$\Rightarrow P(A) \times P(\text{not } B) + P(\text{not } A) \times P(B) = \frac{26}{49}$

$\Rightarrow a \times (1-b) + (1-a)b = \frac{26}{49}$

$\Rightarrow a + b - 2ab = \frac{26}{49} \ldots (\text{i})$

And $P(\text{not } A \text{ and not } B) = \frac{15}{49}$

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$\Rightarrow P(\text{not } A) \times P(\text{not } B) = \frac{15}{49}$

$\Rightarrow (1-a) \times (1-b) = \frac{15}{49}$

$\Rightarrow 1-b-a+ab = \frac{15}{49}$

$\Rightarrow a+b-ab = \frac{34}{49} \ldots$ (ii)

From (i) and (ii),

$a+b = \frac{42}{49} \ldots$ (iii)

and $ab = \frac{8}{49}$

$(a-b)^2 = (a+b)^2 - 4ab = \frac{42}{49} \times \frac{42}{49} - \frac{4 \times 8}{49} = \frac{196}{2401}$

$\therefore a-b = \frac{14}{49} \ldots$ (iv)

From (iii) and (iv),

$a = \frac{4}{7}, b = \frac{2}{7}$

Hence probability of more probable of the two events

$= \frac{4}{7}$

final answer is $49x = 28$.

82. **Ans: 22**

Solution:

There are three cases for this:

(a) Apples are distributed as $(4,\ 3,\ 1)$.

Therefore, total number of ways $= {}^8C_4 \times {}^4C_3 \times 3!$

$= 70 \times 4 \times 6$

$= 1680$

(b) Apples are distributed as $(4,\ 2,\ 2)$.

Therefore, total number of ways $= {}^3C_2 \times {}^8C_4 \times {}^4C_2$

$= 3 \times 70 \times 6$

$= 1260$

(c) Apples are distributed as $(3,\ 3,\ 2)$.

Therefore, total number of ways $= {}^3C_2 \times {}^8C_3 \times {}^5C_3 = 1680$

So, total number of ways including all the cases $= 1680 + 1260 + 1680$

$= 22 \times {}^7P_3$

Therefore, $k = 22$.

83. **Ans: 6**

Solution: $P(A \cap B) = P(A) \cdot P(B)$

$\Rightarrow P(A) = \frac{1}{5}$

Now, $P\left(\frac{\bar{A}}{A \cup B}\right) = \frac{P(\bar{A} \cap (A \cup B))}{P(A \cup B)}$

$\Rightarrow P\left(\frac{\bar{A}}{A \cup B}\right) = \frac{P(B) - P(A \cap B)}{\frac{1}{5} + \frac{1}{2} - \frac{1}{10}} = \frac{\frac{1}{2} - \frac{1}{10}}{\frac{6}{10}} = \frac{2}{3}$

So, $9P\left(\frac{\bar{A}}{A \cup B}\right) = 6$

84. **Ans: 3**

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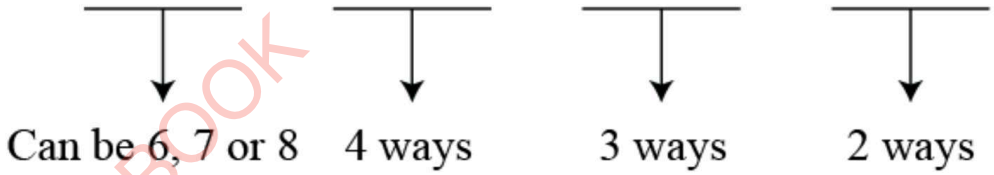
Solution:

$x + \frac{1}{x} = 2 \Rightarrow x = 1 \ldots . \left[x + \frac{1}{x} \geq 2 \text{ as equality occurs only when } x = 1\right]$
$\Rightarrow \sin^{-1} x = \frac{\pi}{2}$

Similarly, $y = -1 \ldots . \left[y + \frac{1}{y} \leq -2 \text{ as equality occurs only when } y = -1\right]$
$\Rightarrow \cos^{-1} y = \pi$
$\sin^{-1} x + \cos^{-1} y = \frac{\pi}{2} + \pi$
$= \frac{3\pi}{2}$
$\Rightarrow m = \frac{3}{2} = 1.5$

85. **Ans: 192**

Solution: Case-1:
Any 5 - digit number > 6000 is all 5 -digits number
Total number > 6000 using 5 -digits $= 5! = 120$
Case-2:
Using 4 -digits



i.e., 3 ways Total number $= 3 \times 4 \times 3 \times 2 = 72$
Total ways $= 120 + 72 = 192$

86. **Ans: 5**

Solution: Total ways is 3^4
Now, $a_1 a_2 = b_1 b_2$
If,
Case $(1) a_1 a_2 = b_1 b_2 = 0$
If at least one of $(a_1$ or $a_2)$ and $(b_1$ or $b_2)$ is "0" i.e. $\left(3^2 - 2^2\right)$ ways
Case (2) $a_1 a_2 = b_1 b_2 = +1$
Both a_1, a_2 is $+1$
or -1
i.e. 2^2 ways
Case (3) $a_1 a_2 = b_1 b_2 = -1$
2^2 ways
Hence, required probability
$= \frac{\left(3^2 - 2^2\right) + 4 + 4}{3^4} = \frac{33}{81} = \frac{11}{27}$

87. **Ans: 3**

Solution: $24^k \rightarrow \left(2^3 \times 3\right)^k$ Exponent of 2 in 13! $\left[\frac{13}{2}\right] + \left[\frac{13}{2^2}\right] + \left[\frac{13}{2^3}\right] = 10$ Exponent of 3 in 13! $\left[\frac{13}{3}\right] + \left[\frac{13}{3^2}\right] = 5$ So $\left(2^3 \times 3\right)^3$ So $\mathrm{K} = 3$

88. **Ans: 2**

Solution:

Let I be the evert that person has infected

P be the evert that test result is positive

$$P\left(\mathrm{I}/\bar{P}\right) = \frac{P(\mathrm{I})\, P(\bar{P}/\mathrm{I})}{P(\mathrm{I})\, P(\bar{P}/\mathrm{I}) + P(\bar{I})\, P(\bar{P}/\bar{I})}$$

$$P(\mathrm{I}) = P\left(\bar{\mathrm{I}}\right) = \frac{1}{2}$$

$$P(P/\mathrm{I}) = \frac{9}{10} \ \& \ P\left(\bar{P}/\bar{I}\right) = \frac{9}{10}$$

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$$P\left(I/\overline{P}\right)=\frac{\frac{1}{2}\times\frac{1}{10}}{\frac{1}{2}\times\frac{1}{10}+\frac{1}{2}\times\frac{9}{10}}=\frac{1}{10}=0.1$$

89. **Ans: 2**

Solution: Let $t=\sin x$

From the graphs of $\sin^{-1}t$ and $\cos^{-1}t$, we know that,

$\sin^{-1}t\geq\cos^{-1}t$ for $t\in\left[\frac{1}{\sqrt{2}},1\right]$

$\Rightarrow\sin x\in\left[\frac{1}{\sqrt{2}},1\right]$

Hence, $x\in\left(\frac{\pi}{2},\frac{3\pi}{4}\right]$

Maximum value is $\frac{3\pi}{4}=\lambda$

So, $\frac{2\lambda}{3}=\frac{\pi}{2}=1.57$

90. **Ans: 7**

Solution:

Case 1

No box is empty

$a+b+c+d+e=10,a,\ b,\ c,\ d,\ e\ \geq1$
Number of solutions $={}^{5+5-1}C_{5-1}={}^{9}C_{4}=126$
Case 2

Exactly 1 box is empty (5 ways)

$a+b+c+d=10,a,\ b,\ c,\ d\ \geq1$

Number of solutions $={}^{6+4-1}C_{4-1}={}^{9}C_{3}=84$
Case 3

Exactly 2 box are empty (6 ways)

$a+c+e=10,a,\ c,\ e\ \geq1$

Number of solutions $={}^{7+3-1}C_{3-1}={}^{9}C_{2}=36$
Case 4

Exactly 3 box are empty (1 way)

$b+d=10,b,\ d\ \geq1$

Number of solutions $={}^{8+2-1}C_{2-1}={}^{9}C_{1}=9$

Therefore, total ways $=126+5\ \times84+6\ \times36+9=771$

Now, $\left[\frac{771}{11}\right]=70$

Hence, sum of digits is 7.

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